

ECON 202-01: Economic Statistics

Fall 2023

Allegheny College

The Business and Economics Department

Class Time: M/W/F, 01:30pm-02:20pm

Classroom: Quigley Hall 217

Instructor: Hang Zhao

Email: hzhao@allegheny.edu

Office hours: W/F 9:00am-11:00am. **Location:** Quigley Hall 118

TH 10:55am-12:10pm, 1:30pm-2:15pm. **Location:** Alden Hall 104

By appointment at <https://calendar.app.google/uS6F298mrJj2hRbEA>

Course Description: The primary purpose of this course is to introduce the basics of statistical concepts, including graphical and numerical descriptive statistics, probability distributions and random variables, sampling distributions, estimation, and tests of statistical significance. The course will also touch upon the basics of simple linear regression, teaching students how to fit a line to a scatter plot. Upon completing the course, students should be proficient in utilizing statistical software like Excel or Stata and be able to interpret the results generated by these platforms.

Distribution Requirements:

The following definitions were taken from the Distribution Requirements: Learning Outcomes website, <https://sites.allegheny.edu/registrar/academic-policies/graduation-requirements/distribution-requirement/distribution-requirements-learning-outcomes/>.

- **Quantitative Reasoning (QR).** Quantitative Reasoning is the ability to understand, investigate, communicate, and contextualize numerical, symbolic, and graphical information towards the exploration of natural, physical, behavioral, or social phenomena.
 - Learning Outcome: Students who successfully complete this requirement will demonstrate an understanding of how to interpret numeric data and/or their graphical or symbolic representations.

Course Objectives:

Course Objectives	Assessment Method
Objective 1, 2, 3, 4 and 5	Homework & Exams

Objective 1: To conduct appropriate statistical analysis of data and interpret statistical results.

Objective 2: (a) to identify vital questions and issues; (b) to collect relevant information and test against relevant criteria and standards; and (c) to defend the inferences made from the information.

Objective 3: To develop clear, concise, audience-appropriate, and grammatically correct written and oral messages.

Objective 4: To apply skills and concepts covered in this course to evaluate the impact of economic problems.

Objective 5: To apply appropriate statistical, mathematical, and technological tools to solve problems or make inferences regarding problems commonly encountered in economics.

Prerequisites: ECON 100, ECON 101, and MATH 140, or MATH 141 or MATH 151.

Required hardware: Hand calculator to be used during exams. A simple version that allows grouping of operations (i.e., using parentheses) and has all the basic math functions will fully suffice. Graphing calculators are also allowed.

Recommended software: Stata (any version after 13 – most basic is fine). You can access Stata on the computers in the lab. You can also purchase a student license, which are available for 6- or 12-month intervals, or perpetually. See the Stata website for pricing:

<https://www.stata.com/order/new/edu/profplus/student-pricing/>

Note: I strongly recommend learning to use Stata along with your peers.

Grading:

- Participation (10%)
- Homework assignments (30%)
- Mid-term exam (30%)
- Final exam (30%)
- Extra Bonus (1%)

Grading Components

- **Homework Assignments.** There will be a total of 4 homework assignments, due via Canvas by 11:59 PM on the given due date. Late assignments will not be accepted. Feel free to collaborate in discussing approaches and answers to the problems in small groups. However, each student should turn in a separate, individually prepared set of answers.

These problem sets will be graded out of 3 points based on competent completion. You will gain 1 point for each of the following: a) 20% of the problems are answered correctly, b) 60% of the problems are answered correctly, c) not skipping *any* portion of *any* problem. Since these scores will not provide a detailed sense of your understanding of the material, you will want to make use of answer keys posted online to assess your performance along the way.

- **Exams.** There will be one mid-term and one final exam. Exams will be administered in person. All work submitted for exams must be yours alone. Discussing exam questions with anyone during the exam period violates academic integrity expectations for this course. **Note that no**

make-up exams will be given and absence from the exams will be recorded as a zero, unless the absence is excused by a documented emergency or religious observance.

Exams will consist of terms and concepts to define/explain, and problems/exercises requiring narrative or quantitative answers.

The midterm will cover materials from topics I-IV. The final exam will focus on material from topics V-VIII but will rely on your understanding of the earlier concepts covered in the course. You may prepare physical notes, not to exceed 3 pages, i.e., 6 sides, to be used during exams.

- **Class Participation.** Regular participation in class activities is essential for your success in this course. Class participation will be based on your active participation in every class through in-class discussion and exercises.
- **Bonus.** Attend office hours at least once this semester to discuss any questions you may have about the class. Even if you don't have specific questions, I'd be happy to hear about your future plans and explore how this course might assist you in achieving them.

Other Learning Resources (optional):

- Textbook (older editions are fine too): Moore, David S., George P. McCabe, Bruce A. Craig. 2021. Introduction to the Practice of Statistics, 10th edition. New York: W.H. Freeman and Company.
- Tutoring: <https://sites.alleggheny.edu/studentsuccess/tutoring/tutor-request/>

Tentative Schedule (subject to change):	Due
Topic I: Descriptive Statistics	
Overview, measures of central tendency, measures of dispersion, standardization, density curves, the normal distribution	
<i>Optional reading: Chapter "Looking at Data – Distributions"</i>	
Week 1, Week 2 & Week 3	Lecture & Exercise
Topic II: Statistical Relationships	
Correlation, the geometry of a line, simple regression, coefficient of determination	
<i>Optional reading: Chapter "Looking at Data – Relationships"</i>	
Week 4 & Week 5	Lecture & Exercise
Topic III. Producing Data	
Sampling, bias, experimental design, randomization	
<i>Optional reading: Chapter "Producing Data"</i>	
Week 6	Lecture & Exercise
10/4	Assignment #1 due by 11:59 PM
Topic IV. Probability	
Rules of probability, random variables, probability distributions	
<i>Optional reading: Chapter "Probability: The Study of Randomness"</i>	
Week 7 & Week 8	Lecture & Exercise

10/11

No class

Topic V: Sampling Distribution

Sampling distribution, Central Limit Theorem

Optional reading: Chapter "Sampling Distributions"

Week 9, Week 10 & Week 11	Lecture, Exercise & Midterm Exam
10/25	Assignment #2 due by 11:59 PM
10/25	Midterm Review
10/27	Midterm Exam - during class period

Topic VI: Inference

Confidence intervals, significance levels, hypothesis testing, p-values

Optional reading: Chapter "Introduction to Inference"

Week 12 & Week 13	Lecture & Exercise
11/17	Assignment #3 due by 11:59 PM

Topic VII: Inference with Distributionst-distribution, t-statistics, inference without knowing σ , standard error of a sample mean, t-test comparing means*Optional reading: Chapter "Inference for Distributions"*

Week 14	Lecture & Exercise
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Topic VIII: Simple Linear Regression

Inference in simple linear regression, regression theory, standard error of the slope

Optional reading: Chapter "Inference for Regression"

Week 15	Lecture, Exercise & Final Exam
12/4	Assignment #4 due by 11:59 PM
12/4	Final Exam Review
Final Exam. TBD, from TBD at TBD.	

1. Statement of Community

- Allegheny students and employees are committed to creating an inclusive, respectful and safe residential learning community that will actively confront and challenge racism, sexism, heterosexism, religious bigotry, and other forms of harassment and discrimination. We encourage individual growth by promoting a free exchange of ideas in a setting that values diversity, trust and equality. So that the right of all to participate in a shared learning experience is upheld, Allegheny affirms its commitment to the principles of freedom of speech and inquiry, while at the same time fostering responsibility and accountability in the exercise of these freedoms.

2. Educational Accommodations (prepared by SASS)

- Students with disabilities who believe they may need accommodations in this class are encouraged to contact Student Accessibility and Support Services (SASS) at (814) 332-2898. Student Accessibility and Support Services is located in Pelletier Library. Please do this as soon as possible to ensure that such accommodations are implemented in a timely fashion. Please see <https://sites.allegheny.edu/studentaccessibility/> for more details.

3. Religious Observance (prepared by SRL)

- If you need to miss class or reschedule a final examination due to a religious observance, please speak to the professor well in advance to make arrangements. Please see <https://sites.allegheny.edu/religiouslife/religious-holy-days/> for more details.

4. Academic Integrity

- Allegheny College has a student-run Honor Code designed to promote individual responsibility and integrity in academic affairs. Each student pledges the following: “I hereby recognize and pledge to fulfill my responsibilities, as defined in the Honor Code, and to maintain the integrity of both myself and the College community as a whole.” Your professor will educate you about the meaning of academic integrity in this course, which may be different in some respects from other courses in which you are enrolled. Please ask the professor if you have questions. Please see <https://sites.allegheny.edu/studentsuccess/academic-honor-program/> for more details.

5. Academic Alerts (prepared by MCSS)

- Allegheny College uses an Academic Alert system to send progress notices to the class deans in the Maytum Center for Student Success (MCSS). **The progress notices are not punitive.** They are an opportunity to connect you with your class dean who can offer additional support and suggest resources if you need assistance. These notices may encourage faculty advisors, coaches (for student-athletes), or other support staff to provide outreach to you as well.

6. Mental Health and Wellness (prepared by Wellness in Student Life; updated July 2023)

- As a college student, there may be times when personal stressors, struggles, and/or traumas interfere with your academic performance and/or negatively impact your daily life. Allegheny College recognizes that mental health is an important piece of the holistic human experience, and that this experience influences your academic success. We encourage students to prioritize their mental well-being by seeking services and support as needed.
- Keep in mind, course deadlines, absences, and accommodations for mental health are subject to the course policies and expectations that are set within this syllabus. Students are encouraged to communicate with their professors as soon as possible regarding their needs and seek support if their mental health impacts their academic performance or daily life. When you find yourself struggling emotionally, remember that there are supports available, and you are not alone.
- Students who are in need of mental health support can access multiple resources to receive support. The 98.6 CampusCare app provides physical and behavioral health consultation and supports free of charge 24/7 via telehealth for any actively enrolled student who downloads and signs up with their smartphone. Allegheny College has also partnered with ECPS (Edinboro Counseling and Psychological Services) to provide in-person therapy on campus. The ECPS partnership with Allegheny College delivers holistic mental health services to the Allegheny College student community through brief individual counseling, and crisis support. Students may request services from ECPS by calling (814) 332-4368 or by emailing the Wellness Case Manager at wellness@allegheny.edu.
- If you or someone you know is experiencing a mental health emergency, please call the 988 Suicide and Crisis Lifeline, Public Safety (814-332-3357), or 911. All of these urgent resources are available 24/7/365. *When in doubt, reach out.*

7. Gator Success Grants

- The [Gator Success Grant program](#) is designed to provide students with additional financial assistance to offset the total cost of college attendance and to encourage successful on-time degree completion. Currently enrolled students who have an unanticipated need are invited to apply. Grants are usually from a few hundred to a few thousand dollars, depending on the request and the student's need. Please direct questions to studentsuccess@allegheny.edu.

8. Writing and Speaking Consultants (prepared by the Directors of Writing and Speaking)

- Experienced peer writing and speaking consultants in the Maytum Center for Student Success help writers and speakers make academically responsible choices and determine strategies for effective communication at any stage in the writing or speaking process, for assignments in any discipline, or for writing and speaking situations unrelated to coursework (e.g., personal statements, cover letters, interviews, conference presentations). Both appointments and drop-in sessions are available. To view the hours of operation, and to make an appointment, visit the [Maytum Center for Student Success website](#).

9. Gator Day / Scholar Symposium (prepared by MCSS and URSCA)

- In each semester, we have one day set aside for programming, with no classes. In the Fall (usually late October), **Gator Day** features programs and workshops that highlight educational opportunities at Allegheny, that prepare students for life after graduation, and that help to create a greater sense of campus community. In the Spring, the **Richard J Cook – Teresa M Lahti Scholars Symposium** (the day immediately after classes end and before exams begin), we gather to celebrate student research, scholarship, and creative activities on campus. We encourage all students to participate.

10. Class Deans (prepared by MCSS)

- The [Office of Class Deans](#) in Pelletier Library serves as a place for students to begin seeking assistance and answers to their questions related to college life. If you have a question and don't know who to ask – ask a class dean. If you want advice – talk to a class dean. Niki Fjeldal is the First Year Class Dean, Amy Stearns is the Second Year Class Dean / Director of Transfer Advising, and Jonathon May is the Third/Fourth Year Class Dean. Please contact them individually or through studentsuccess@allegheny.edu.

11. Recording Policy (prepared by Campus Life and Community Standards Committee)

- In remote teaching and learning contexts, there is an understanding that instructors (including faculty, coaches, staff, and facilitators) have the right to record class and discussion proceedings. All other participants must request permission of the instructor in advance before making any additional recordings. An instructor's pre-recorded material, including lectures and class notes, may not be shared with anyone outside of the course's current enrollment without the consent of the instructor. Similarly, under no circumstances may a recording, still, screenshot, picture, or any other media of any sort be altered or circulated by anyone outside of its original intent. No content may be distributed outside the circle of participants without the consent of all who appear or are heard. The college community is reminded of Allegheny College's Recording Policy, outlined in 9.8 of the *Faculty Handbook*, as well as Allegheny's Statement of Community. Violations will be referred to the Student Conduct system. Classes may also be recorded if approved as an educational accommodation through the Student Accessibility and Support Services. Faculty are further reminded that recordings of class activities are educational records protected under FERPA. While students

do not have a right to be anonymous in a class in which they are enrolled, they have the legal right to be anonymous to third parties. Please direct questions about FERPA to the Registrar.

12. Harassment, Discrimination, and Sexual Misconduct Reporting (prepared by the Title IX Coordinator)

- Allegheny College and Federal law (Title IX) prohibit discrimination and harassment on the basis of sex in College's programs and activities. If you, or someone you know, has experienced sex-based discrimination or harassment, sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, or retaliation contact the College's Title IX Coordinator to make a report and/or access support and resources:

Gilly Ford, Title IX Coordinator

814-332-3085 (direct line), gford@allegheny.edu

- *You can access support and resources even if you do not want to take any further action.* The College respects your right to file or not file a formal complaint. The Title IX Coordinator must take action only in situations that pose a serious threat of harm to you or the community.

Notice that Faculty are Responsible Employees with Mandatory Reporting Obligations:

- Unless specifically identified otherwise by policy, all faculty, staff, and administrators of Allegheny College are considered mandated reporters and are required to notify the College's Title IX Coordinator of any incident of discrimination and harassment disclosed to them, or known to them, involving a member of the campus community. Mandated reporting helps to ensure that individuals who are experiencing or have experienced discrimination or harassment are connected to the full range of resources and options afforded to them.
- If you are in need of support or advocacy and wish to discuss such matters confidentially, you are invited to consult any of the resources listed below. Unlike other College employees, the options below are *confidential* and are exempt from mandatory reporting obligations to the Title IX Coordinator. These resources are free and include both internal and external options.

On Campus Options:

- Meadville Medical Center @ Winslow - (814-332-4355) - Walk-ins welcome - Open from M-F 10am-3pm (last patient check-in at 2:30pm) while classes are in session.
- ECPS partnership with Allegheny College (Edinboro Counseling and Psychological Services) - (814) 332-4368 - Please call to schedule an appointment.
- 98point6 CampusCare - Allegheny students can download the app for iOS or Android smartphones, confirm their cell phone number, and complete their profile to receive services. This physical/behavioral telehealth service is free of charge for all actively enrolled students at Allegheny College.
- Wellness in Student Life - wellness@allegheny.edu - (814) 332-2782

Off Campus Options:

- [Meadville Medical Center Emergency Room](#), 751 Liberty Street – (814) 333-5500
- [Meadville Crisis Line](#), (814) 724-2732

- [Women's Services, Inc.](#), (814) 724-4637 (office)
- 24- hour Hotlines: (814) 333-9766 or (888) 881-0189
- If you would like more information on the College's Harassment, Discrimination and Sexual Misconduct Policy click [here](#).
- If you would like to access the College's brochure summarizing your rights and resources, click [here](#).